Gut Microbiome Signatures of Obesity

Statistical Testing, Classical ML, and Phylogenetic Deep Learning

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Abstract

Obesity is associated with substantial inter-individual variability in gut microbiome composition, but translating that variability into robust, clinically useful signals remains challenging because microbiome data are sparse, high-dimensional, and highly skewed. We analyzed the LeChatelier et al. cohort and framed the task as binary classification of Lean (BMI <= 25) vs Obese (BMI >= 30), while also identifying taxa associated with group differences.

Our pipeline combines preprocessing with MIPMLP, non-parametric statistical testing with false-discovery-rate correction, and predictive modeling with Logistic Regression, Random Forest, and iMic-based CNN models. In this run, five taxa passed FDR < 0.1; the strongest signal was Fretibacterium_fastidiosum (p = 1.20e-05), followed by Clostridium_sp_CAG_58 (p = 6.12e-05). A complementary taxonomy-aware mi-Mic test was also strongly significant (nested p = 5.80e-59), reported 47 path/leaf detections, and recovered the same two leading taxa at the leaf level. On hold-out evaluation, Logistic Regression achieved AUC 0.700 (F1 0.738), Random Forest achieved AUC 0.629 (F1 0.623), and iMic achieved AUC 0.783 with recall-oriented thresholding.

Overall, the results support a reproducible end-to-end workflow and show that phylogenetic image representations can improve discrimination beyond tabular baselines in this dataset. We also highlight the operating-point trade-off between sensitivity and specificity for screening-oriented deployment. Model differences are reported descriptively because no formal significance test between models was completed in this submission.

1. Introduction

The gut microbiome is increasingly linked to obesity and metabolic disease, but extracting stable signal is difficult: microbiome matrices include many zeros, compositional constraints, and correlated taxa. This project asks two connected questions: (1) which taxa are significantly associated with Lean vs Obese phenotypes, and (2) whether predictive models using microbiome features, including phylogenetic image encoding, improve obesity classification.

We use a story-driven workflow: characterize the cohort, test statistical hypotheses, build interpretable tabular baselines, then evaluate a deep-learning model that preserves phylogenetic structure.

2. Methods

Data source: LeChatelier cohort tables in this repository. After BMI filtering (Lean <= 25, Obese >= 30), the final cohort was n=265 (Lean=98, Obese=167).

Preprocessing: MIPMLP taxonomy aggregation (species-level), relative normalization, rare taxa filtering, then global log transform (log10(X+1e-6)).

Statistical analysis: per-taxon Mann-Whitney U tests with Benjamini-Hochberg FDR correction (q < 0.1), plus a taxonomy-aware mi-Mic test on the same post-MIPMLP species table. In this cohort, the paper-style sub-PCA merge collapsed the data to one synthetic feature, so mi-Mic was run on the preserved species table X instead.

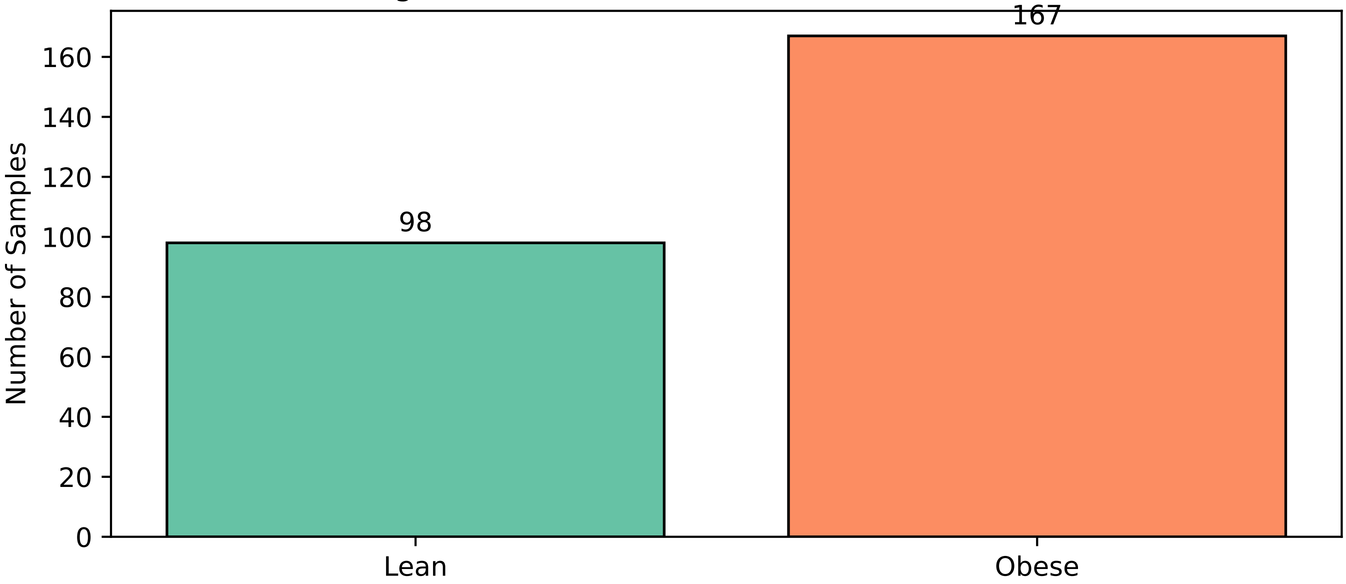
Models: L1-regularized Logistic Regression, Random Forest with scaler+SMOTE+grid-search, and iMic CNN optimized with Optuna.

Evaluation: fixed shared hold-out split (80/20), fixed seeds, training-only tuning for the iMic validation step, and hold-out reporting shared across all primary models.

3. Results: Data Profile, Statistical Signals, and mi-Mic

- Most significant taxon: Fretibacterium_fastidiosum (p = 1.20e-05)
- Second most significant taxon: Clostridium_sp_CAG_58 (p = 6.12e-05)
- Total significant taxa after FDR correction (q < 0.1): 5
- mi-Mic nested a priori test: p = 5.80e-59 (strong taxonomic structure).
- mi-Mic significant path/leaf detections: 47.
- mi-Mic overlap with the leaf-level screen: Fretibacterium_fastidiosum and Clostridium_sp_CAG_58.

Figure 1. Class Distribution in Filtered Cohort



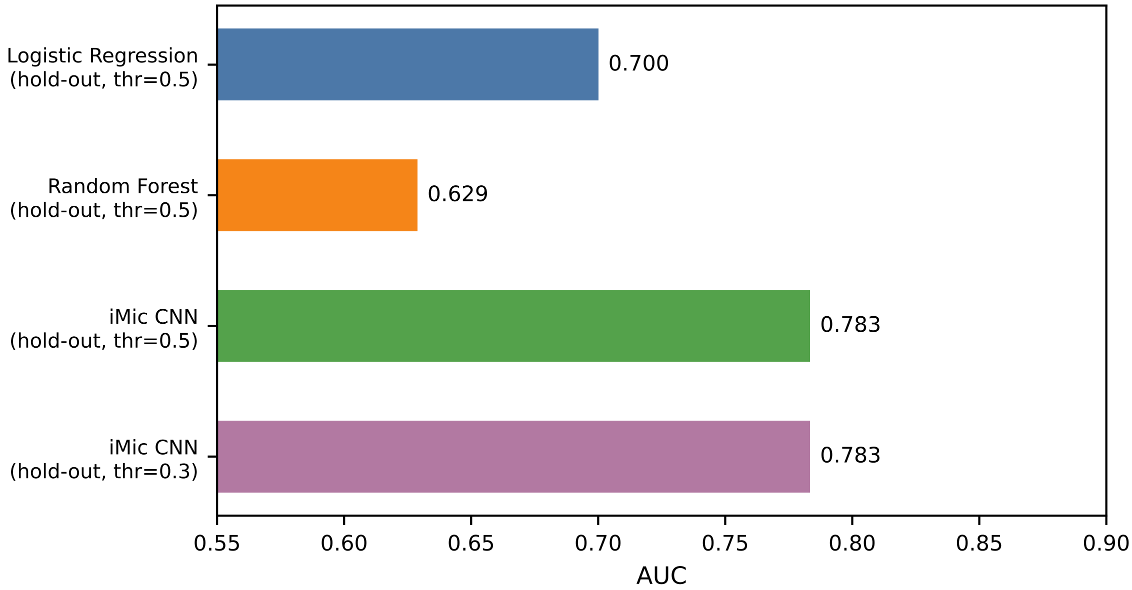
Interpretation: The cohort is imbalanced toward Obese samples. This supports using class-aware training choices(balanced logistic regression and SMOTE in RF), and explicit threshold analysis during evaluation.

3. Results: Predictive Modeling Performance

Table 1. Hold-out Performance Summary (Latest Run)

Model	AUC	Accuracy	Precision	Recall	F1
Logistic Regression (hold-out, thr=0.5)	0.7	0.679	0.75	0.727	0.738
Random Forest (hold-out, thr=0.5)	0.629	0.566	0.679	0.576	0.623
iMic CNN (hold-out, thr=0.5)	0.783	0.679	0.667	0.97	0.79
iMic CNN (hold-out, thr=0.3)	0.783	0.698	0.673	1.0	0.805

Figure 2. AUC Comparison Across Models



3. Results: Operating-Point Calibration

Figure 3. Confusion Matrix at iMic Threshold = 0.3

At threshold 0.3, iMic reaches recall = 1.00 for Obese cases with precision = 0.673 and accuracy = 0.698.

The confusion matrix (n=53) shows TP=33, FN=0, TN=4, FP=16.

Storytelling takeaway: lowering the threshold shifts the model toward sensitivity (screening behavior), reducing missed Obese cases while increasing false positives among Lean samples. The suitable threshold depends on downstream clinical cost.

4. Discussion

This analysis follows a coherent path from biological question to deployable model behavior. Statistically, five taxa showed significant group differences after FDR control. Predictively, tabular models established a baseline (LR outperforming RF in this run), while iMic improved discrimination and demonstrated stronger sensitivity at tuned thresholds.

The mi-Mic analysis strengthens the missing bridge between statistics and modeling: its nested a priori test was highly significant (p = 5.80e-59), it reported 47 taxonomically structured detections, and its leaf-level overlap with the flat Mann-Whitney screen consisted of the same two headline taxa, Fretibacterium_fastidiosum and Clostridium_sp_CAG_58.

The strongest practical signal is not a single metric but the shape of trade-offs: iMic can be operated as a high-sensitivity screening model when missing Obese cases is costly. The main methodological limitation is that no formal significance test was completed between model performances, so observed score differences should be interpreted descriptively.

4.1 Biological Interpretation of the Two Leading Taxa

Fretibacterium_fastidiosum was higher in Lean participants. A cautious mechanistic reading is possible here: the species was described as asaccharolytic, with acetate and propionate among its metabolic end products, and members of the broader Synergistota/Synergistetes lineage are known amino-acid fermenters. Therefore, in our dataset, this taxon can be discussed as a marker of a peptide- and amino-acid-fermenting ecological niche rather than a sugar-driven one. This interpretation is indirect, but it is biologically compatible with nutrition literature showing that higher-protein diets can enhance fullness or satiety. Importantly, our data contain no dietary intake measurements, so we cannot claim that higher protein intake caused either the increase in Fretibacterium_fastidiosum or the lean phenotype (Vartoukian et al., 2013; Geng et al., 2022; de Carvalho et al., 2020).

Clostridium_sp_CAG_58 was enriched in Obese participants. Here the scientifically safest wording is more conservative: Clostridium sp. CAG:58 is currently an unclassified placeholder taxon in NCBI, with limited species-level functional characterization. We therefore interpret it as an obesity-associated clostridial signal in this cohort, not as proof that this organism specifically prefers simple carbohydrates or directly drives carbohydrate craving. More generally, gut microbial products can modulate appetite-regulatory and gut-brain signaling pathways, but attributing such a mechanism specifically to CAG:58 would go beyond the available evidence (NCBI Taxonomy Browser, taxid 1262824; Fetissov, 2017).

5. Limitations

- Single cohort; no external validation cohort in this submission.

- Cross-sectional design supports association, not causality.